

Confidently Wrong on Unseen Domains: Domain-Adversarial Training Silently Disables Out-of-Distribution Monitoring

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Abstract

Background. Domain-adversarial invariance training is widely recommended for removing acquisition shortcuts from medical imaging models. Its effect on the out-of-distribution (OOD) monitoring that such models rely on at deployment has not been measured.

Methods. We trained a dual-encoder factorization model on ISIC 2019 (dermoscopy, 25,331 images) and PAD-UFES-20 (clinical photographs, 2,298 images), sweeping the adversarial weight λ_{adv} across seven values with three to five seeds each (27 runs). At every λ_{adv} we measured domain leakage by linear probe, in-distribution balanced accuracy, calibration on in-distribution and out-of-domain data, OOD detection with five scoring rules, and zero-shot cross-domain diagnostic accuracy on a patient-level held-out PAD split. We validated on Fitzpatrick17k, a third dataset never seen by the adversary.

Results. Domain leakage fell to its chance floor at the smallest non-zero weight tested (balanced accuracy $0.915 \rightarrow 0.553$, floor 0.5), while in-distribution balanced accuracy did not degrade ($0.692 \rightarrow 0.707$). Over the same range, OOD detection fell from AUROC 0.863 to 0.508 and inverted below chance (0.413), with OOD samples ranked as more in-distribution than in-distribution data throughout the score distribution rather than only in the tails. Cross-domain diagnostic accuracy did not improve at any λ_{adv} (balanced accuracy 0.249–0.291 against a six-class floor of 0.167, and below the 0.298 of a non-adversarial control at every setting), and predictions collapsed toward the benign majority class: actinic keratosis recall fell from 0.244 to 0.016 ($n = 202$). Mean confidence on never-seen-domain images rose from 0.489 to 0.930, overtaking in-distribution confidence, so that out-of-domain calibration error tripled (ECE $0.247 \rightarrow 0.746$) while in-distribution ECE stayed flat at ≈ 0.10 . Leakage did not mediate the OOD collapse: it saturated at chance while AUROC kept falling, and at the backbone leakage was flat while AUROC fell $0.749 \rightarrow 0.475$. Six candidate explanations were tested and refuted.

Conclusions. When domain-adversarial training succeeds, models become confidently wrong on unseen domains and their OOD monitoring fails, while in-distribution metrics improve. OOD detection and out-of-domain calibration should be reported as a function of achieved invariance rather than at a single operating point.

Keywords: domain-adversarial training; out-of-distribution detection; calibration; distribution shift; skin lesion classification; model monitoring

1 Introduction

Deep models for skin lesion classification are known to exploit acquisition artefacts rather than pathology: ruler marks, ink annotations, imaging hardware and site identity all correlate with diagnosis in public benchmarks [1, 2, 3]. The standard remedy is to make the representation invariant to the acquisition domain, most often through domain-adversarial training with a gradient reversal layer [4, 5]. The stated purpose of this remedy is safe deployment across sites.

Deployment across sites also relies on a second mechanism: out-of-distribution monitoring. A deployed system is expected to flag inputs unlike its training data so that a human reviews

them rather than the model acting alone. This paper asks what invariance training does to that mechanism.

The answer is that it destroys it, and does so invisibly. Sweeping the adversarial weight in a controlled setting, we find that OOD detection collapses from AUROC 0.863 to 0.508 at the smallest non-zero weight tested and then inverts below chance, so that images from an unseen clinic are ranked as *more* routine than familiar ones. Over the same range the model becomes more confident on those unseen images than on its own held-out test data, and its predictions collapse toward the benign majority class. None of this is visible to the metrics a developer monitors: in-distribution accuracy does not fall, in-distribution calibration does not change, and the domain leakage probe — the very metric the technique is adopted to improve — improves as intended.

The irony is the point. The technique is adopted for cross-site robustness and it degrades precisely the cross-site safety machinery, while every monitored number moves in the direction its user is hoping for.

Contributions.

1. A controlled intervention: a seven-point sweep of the adversarial weight with 27 training runs in a fixed architecture, isolating the effect of invariance pressure from architectural and data confounds.
2. The safety failure: OOD detection collapses as a cliff rather than a trade-off curve, inverts below chance, and does so on a third domain the adversary never saw.
3. The invisibility: in-distribution accuracy, in-distribution calibration and the domain leakage probe all fail to reveal it.
4. Evidence that the leakage probe does not mediate the collapse, so it cannot serve as a proxy for it.
5. Six candidate explanations tested and refuted, plus a pre-registered synthetic reproduction that failed — delimiting the phenomenon for future work.
6. A reporting protocol that makes the failure visible at near-zero cost.

2 Related work

Shortcut learning in dermatology. Winkler et al. [1] showed that ink markings shift melanoma predictions; Bissoto et al. [2] showed that classifiers retain competitive accuracy on ISIC data when the lesion itself is masked; Daneshjou et al. [3] documented degraded performance on darker skin tones. These motivate invariance-based remedies.

Invariant representation learning. Domain-adversarial training [4] and domain separation networks [5] remove domain information by adversarial or orthogonality constraints. **The architecture used here is equivalent to a domain separation network and is not a contribution of this paper.** It is a vehicle for the intervention.

Covariate shift versus semantic novelty. Full-spectrum OOD detection [6] argues that the OOD literature conflates two distinct shifts, and proposes a four-way taxonomy: training ID, *covariate-shifted ID*, near-OOD and far-OOD. Crucially, it prescribes that covariate-shifted in-distribution samples should be **tolerated as in-distribution rather than flagged**, on the grounds that rejecting them contradicts the goal of generalization. OpenOOD v1.5 [7] adopts the same full-spectrum setting at ImageNet scale and for foundation models.

Our setting falls squarely in the covariate-shifted-ID cell: PAD-UFES-20 contains the same diagnostic classes as ISIC, photographed differently. A reader of [6] might therefore regard a model that accepts these images as in-distribution as behaving *correctly*.

We argue this is exactly the gap. **Tolerance of covariate shift is safe only when accompanied by generalization, and the two are evaluated separately or not at all.** Our models achieve the prescribed tolerance — by $\lambda_{\text{adv}} = 2$ they rank out-of-domain images as more in-distribution than their own test data — while cross-domain balanced accuracy sits at 0.249 against a floor of 0.167, plain accuracy is below the majority-class rate, and confidence on those images exceeds in-distribution confidence. Tolerating inputs one cannot classify is worse than flagging them, and no current protocol separates the two cases. We therefore report acceptance and cross-domain accuracy jointly throughout.

Calibration as a route to invariance — and the reverse. Ovadia et al. [8] showed that calibration degrades under shift for ordinary training. Wald et al. [9] went further and established the forward direction: enforcing *multi-domain calibration* yields invariance, with models so calibrated provably free of certain spurious correlations, and they propose a calibration loss (CLOvE) validated on WILDS.

Our results show the reverse implication does not hold. Obtaining invariance adversarially does not yield calibration; it *destroys* out-of-domain calibration in a dose-dependent way while leaving in-distribution calibration untouched. Calibration and adversarial invariance are therefore not interchangeable routes to the same representation, and the direction of enforcement matters.

What is new here. (i) The *shape* of the dependence — a cliff at the smallest effective adversarial weight rather than a graded trade-off; (ii) *sub-chance* inversion, in which out-of-domain data is ranked as more typical than in-distribution data throughout the score distribution; (iii) the demonstration that the failure is invisible to every routinely monitored metric; (iv) the finding that the standard domain leakage probe neither mediates nor predicts it; (v) the consequence for calibration and for diagnostic decisions on out-of-domain data; and (vi) the observation that full-spectrum tolerance is achieved here without the generalization that would make it safe.

3 Materials and methods

3.1 Data

ISIC 2019 [10]: 25,331 dermoscopic images, eight diagnostic classes (MEL, NV, BCC, AK, BKL, DF, VASC, SCC), CC BY-NC 4.0. **PAD-UFES-20** [11]: 2,298 smartphone clinical photographs, six classes mapping into the same label space (DF and VASC absent), CC BY 4.0. **Fitzpatrick17k** [12]: 3,887 of 16,577 images obtained; annotations CC BY-NC-SA 3.0, images web-scraped under mixed copyright, used for evaluation only and not redistributed.

Images were cropped to a soft lesion-centred bounding box via Otsu thresholding on the deviation from the border median, expanded by a 0.30 margin with a 0.70 minimum crop ratio, and resized to 224×224 .

3.2 Splits and the PAD held-out set

ISIC was split 60/20/20 into train/validation/test, stratified by label (`random_state = 42`).

PAD appears in training and would otherwise also be the OOD set. The classification loss never sees PAD labels (`ignore_index = -1`), but PAD images enter the context and adversarial branches. To remove the resulting circularity we partitioned PAD *at patient level* into `pad_adv` (1,582 images, 961 patients), visible to those branches during training, and

pad_heldout (716 images, 412 patients), never seen in any form. No patient and no lesion appears in both. **All OOD and cross-domain numbers reported in the main text use pad_heldout.** Fitzpatrick17k provides a second never-adversarial domain.

3.3 Architecture and objective

A dual-encoder model with EfficientNet-B3 backbones produces a lesion latent $z_\ell \in \mathbb{R}^{16}$ (batch-normalised, denoted z_ℓ^{norm}) and a context latent $z_c \in \mathbb{R}^{64}$. Diagnosis uses z_ℓ^{norm} only, through a linear head. The objective is

$$\mathcal{L} = \mathcal{L}_{\text{cls}} + \lambda_{\text{ctx}} \mathcal{L}_{\text{ctx}} + \lambda_{\text{adv}} \mathcal{L}_{\text{adv}} + \lambda_{\text{orth}} \mathcal{L}_{\text{orth}}, \quad (1)$$

with $\mathcal{L}_{\text{cls}}$ on ISIC only, $\mathcal{L}_{\text{ctx}}$ a domain cross-entropy on z_c , $\mathcal{L}_{\text{adv}}$ a domain cross-entropy on z_ℓ^{norm} behind a gradient reversal layer, and $\mathcal{L}_{\text{orth}}$ a squared-cosine penalty between the two latents ($\lambda_{\text{orth}} = 1$ throughout).

AdamW, base learning rate 10^{-4} , adversary learning rate multiplier 30, classifier multiplier 0.2, 40 epochs, batches balanced across domains, checkpoint selected on validation accuracy.

3.4 The intervention

$\lambda_{\text{adv}} \in \{0, 0.25, 0.5, 1, 2, 4, 8\}$; five seeds at $\lambda_{\text{adv}} \in \{0, 2, 8\}$ and three at the interior points, for 27 runs. Everything else is held fixed, so λ_{adv} is the only manipulated variable.

3.5 Measurements

Domain leakage. Logistic probe on frozen features predicting the source dataset, reported as **balanced accuracy, whose chance floor is 0.5 for this two-domain problem**. Earlier work on this dataset pair reports plain accuracy against a majority floor of 0.688; the two scales must not be compared to each other. Both are stated where relevant.

OOD detection. MSP, free energy, maximum cosine to a class centroid, class-conditional Mahalanobis with shared covariance, and k -NN distance ($k \in \{10, 50, 200\}$). All statistics are fit on the ISIC training split alone. **These five rules are not independent evidence:** the classifier is a linear map from z_ℓ^{norm} , so all five are functions of the same 16-dimensional vector. Independent evidence comes from repeating the analysis on the pre-projection backbone representation (Section 4.5).

Calibration. Expected calibration error, 15 bins, computed separately on ISIC test and on pad_heldout restricted to the six shared classes. **Note that out-of-domain ECE here is arithmetically close to mean confidence minus accuracy**, so the confidence, accuracy and ECE results in Section 4.4 are three views of one phenomenon, not three independent findings.

Cross-domain diagnosis. Zero-shot evaluation on pad_heldout over the six shared classes. PAD labels were never used for $\mathcal{L}_{\text{cls}}$.

4 Results

4.1 The intervention

Table 1 reports every measurement as a function of λ_{adv} , the only manipulated variable. Figure 1 shows the four quantities that matter most on a shared axis.

Read across a single row and nothing is wrong. Read down the columns and two of them improve while two collapse.

Table 1: The intervention. Mean $\pm$ s.d. across seeds. Leakage is balanced accuracy (chance 0.5); cross-domain accuracy is balanced over the six classes shared with PAD-UFES (chance 0.167). Arrows mark the direction a practitioner would call favourable.

λ_{adv}	n	Leakage $\downarrow$	ID bal. acc $\uparrow$	ID ECE	OOD ECE	OOD AUROC	Fitz. AUROC	X-dom bal. acc
0	5	0.915 ± 0.021	0.692	0.107	0.247	0.863 ± 0.025	0.764 ± 0.038	0.291 ± 0.016
0.25	3	0.553 ± 0.008	0.696	0.101	0.684	0.508 ± 0.022	0.439 ± 0.061	0.291 ± 0.020
0.5	3	0.555 ± 0.029	0.701	0.096	0.724	0.449 ± 0.008	0.449 ± 0.025	0.270 ± 0.034
1	3	0.569 ± 0.010	0.701	0.097	0.733	0.439 ± 0.040	0.457 ± 0.088	0.261 ± 0.016
2	5	0.584 ± 0.027	0.707	0.099	0.746	0.413 ± 0.021	0.391 ± 0.068	0.249 ± 0.030
4	3	0.548 ± 0.035	0.686	0.100	0.726	0.443 ± 0.038	0.431 ± 0.036	0.276 ± 0.025
8	5	0.625 ± 0.045	0.679	0.099	0.713	0.481 ± 0.026	0.474 ± 0.060	0.269 ± 0.023
EffNet-B3 control		0.802	0.683	0.096	—	0.726 ± 0.021	0.649	0.298 ± 0.010
ResNet-50 baseline		0.979	0.655	0.101	—	0.868 ± 0.019	0.994	0.391

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Figure 1: **The intervention, four views, one axis.** Leakage falls to its chance floor and in-distribution balanced accuracy does not degrade — both favourable. Over the same range OOD AUROC collapses through chance and inverts, and out-of-domain calibration error triples. A developer monitoring the first two panels would see nothing wrong.

4.2 Invariance is achieved, and in-distribution metrics do not warn

Domain leakage falls from 0.915 to 0.553 between $\lambda_{\text{adv}} = 0$ and $\lambda_{\text{adv}} = 0.25$, against a chance floor of 0.5. By the smallest non-zero weight tested, the lesion representation is very nearly uninformative about the acquisition source: the intended effect of the method, obtained essentially in full.

It is obtained without an in-distribution cost. Balanced accuracy on the ISIC test split moves from 0.692 to 0.707 over the useful range — we claim no improvement, the difference being small relative to seed variance — and in-distribution expected calibration error is flat at ≈ 0.10 at every λ_{adv} . A practitioner tracking accuracy, in-distribution calibration and the leakage probe would conclude that the method had worked exactly as advertised.

4.3 Out-of-distribution detection collapses, and inverts

Over the same interval, Mahalanobis AUROC against the held-out PAD partition falls from 0.863 to 0.508, and reaches 0.413 at $\lambda_{\text{adv}} = 2$.

Two features of this collapse matter more than its size.

It is a cliff, not a trade-off curve. The transition is essentially complete at the smallest non-zero weight tested. Between $\lambda_{\text{adv}} = 0.5$ and $\lambda_{\text{adv}} = 8$ the value moves only between 0.413 and 0.481, with no ordering that would let a practitioner tune for a safe compromise. There is no operating point on this curve at which invariance has been obtained and detection survives.

It goes below chance. At $\lambda_{\text{adv}} = 2$ the detector ranks images from a clinic the model has never seen as *more* in-distribution than images from its own test split. This is not a tail artefact. Removing the pooled 90th, 95th or 99th percentile of scores leaves AUROC at 0.412, 0.416 and 0.424 respectively against 0.427 for the full set, and the medians themselves invert (ISIC 13.8 versus `pad_heldout` 10.0). The shift is distributional throughout.

All five scoring rules move together — MSP $0.970 \rightarrow 0.437$, energy $0.983 \rightarrow 0.441$, cosine $0.901 \rightarrow 0.435$, k -NN $0.941 \rightarrow 0.424$ (stable across $k \in \{10, 50, 200\}$). As noted in the methods these are five views of one 16-dimensional representation and are not independent evidence; the independent replication is at the backbone (Section 4.5) and in the single-encoder model (Section 4.7).

The collapse is not a numerical artefact of the covariance estimate. Ledoit–Wolf shrinkage leaves the $\lambda_{\text{adv}} = 2$ value unchanged at 0.413, and effective dimensionality barely moves across the sweep (participation ratio $4.879 \rightarrow 4.500$ while leakage falls $0.914 \rightarrow 0.553$).

It is not confined to the domain the adversary was trained against. On Fitzpatrick17k, absent from every training branch, AUROC falls $0.764 \rightarrow 0.391$, while a non-adversarial baseline reaches 0.994 on the same images.

4.4 The model becomes confidently wrong on unseen domains

At $\lambda_{\text{adv}} = 0$ the model is appropriately hesitant on unfamiliar data: mean maximum-softmax confidence is 0.915 on the ISIC test split and 0.489 on `pad_heldout`. By $\lambda_{\text{adv}} = 0.25$ that gap has closed (0.913 versus 0.898), and by $\lambda_{\text{adv}} = 2$ it has inverted — the model is *more* confident on images from a clinic it has never seen (0.930) than on its own held-out test set (0.910).

Confidence rises while accuracy falls. Cross-domain diagnostic accuracy does not improve at any λ_{adv} : balanced accuracy spans 0.249–0.291 against a six-class floor of 0.167, no setting exceeds the 0.291 obtained at $\lambda_{\text{adv}} = 0$, and every setting is below the 0.298 of the non-adversarial EfficientNet-B3 control. Plain accuracy falls $0.245 \rightarrow 0.184$, beneath the 0.385 majority-class rate throughout.

Macro AUC is roughly preserved ($0.605 \rightarrow 0.581$), so discrimination survives while the decision rule does not. Predictions collapse toward the benign majority class: nevus predictions rise from 113 to 374 of 716 images, and **actinic keratosis recall falls from 0.244 to 0.016** ($49/202 \rightarrow 3/202$ images, mean over five seeds). Squamous cell carcinoma moves the same way ($0.088 \rightarrow 0.021$, $6/66 \rightarrow 1/66$) but the count is too small to carry a claim on its own, and melanoma ($n = 9$) supports none.

The resulting out-of-domain calibration error triples, from 0.247 to 0.746, while in-distribution ECE stays flat at ≈ 0.10 . We stress that confidence, accuracy and calibration here are three views of a single phenomenon rather than three independent results: in this setting out-of-domain ECE is arithmetically close to mean confidence minus accuracy.

4.5 Leakage does not mediate the collapse

The natural hypothesis is that removing domain information is what breaks detection. Three results rule it out.

First, leakage **saturates at its chance floor by** $\lambda_{\text{adv}} = 0.25$ (0.553 against a floor of 0.5) while AUROC continues to fall from 0.508 to 0.413 as λ_{adv} rises further. The damage continues after invariance is complete.

Second, at the pre-projection backbone, leakage is essentially flat across the whole sweep ($0.982 \rightarrow 0.941$) while backbone Mahalanobis AUROC falls $0.749 \rightarrow 0.475$. Domain information remains linearly decodable there, and detection collapses anyway. Because the backbone is a different representation from the 16-dimensional latent, this is the independent replication that the five-detector agreement does not provide.

Table 2: Invariance objectives other than gradient reversal, single-encoder backbone, 3 seeds per cell. Leakage is balanced accuracy, floor 0.5. **IRM and GroupDRO use PAD labels** through their environment and group risks; the remaining objectives do not, so their cross-domain numbers are not comparable and the two regimes are kept in separate blocks.

Objective	Weight	Leakage	Maha. AUROC	ID bal. acc	OOD ECE
<i>No access to PAD labels</i>					
ERM	—	0.973	0.838	0.761	—
CORAL	100	0.998	0.999	0.777	—
MMD	1	0.967	0.813	0.779	—
MMD	100	0.900	0.716	0.730	—
MMD	3000	0.893	0.748	0.382	0.142
MMD	10000	0.836	0.665	0.337	0.267
GRL	$\lambda_{\text{adv}} = 0.25$	0.763	0.525	0.845[†]	—
<i>Uses PAD labels — not comparable on cross-domain accuracy</i>					
IRM	100	0.950	0.729	0.167	—
GroupDRO	10	0.976	0.868	0.783	—

[†] plain accuracy for the GRL row; see Table 3.

Third, and most directly, two training objectives traverse the same leakage range with opposite consequences (Section 4.6). Leakage alone therefore does not determine the outcome.

We accordingly report leakage as a measured quantity and never as the explanatory variable, and plot all results against λ_{adv} rather than against leakage.

4.6 The failure is specific to adversarial optimisation

Adversarial training is one route to invariance among several. Table 2 sweeps four alternatives on the single-encoder backbone.

Only MMD moves leakage appreciably, and it does so at a price gradient reversal never pays. Escalating the MMD weight from 1 to 10,000 buys 0.13 of leakage ($0.967 \rightarrow 0.836$, floor 0.5) at a cost of 0.44 of in-distribution balanced accuracy ($0.779 \rightarrow 0.337$), and leaves every detector well above chance (Mahalanobis 0.665, k -NN 0.738, energy 0.694 at the highest weight). IRM reaches the extreme of the same pattern: at weight 100 its balanced accuracy collapses to 0.167 while leakage remains at 0.950. CORAL does not induce invariance at all; at weight 100 it makes the representation *more* domain-separable (leakage 0.998). GroupDRO leaves leakage unmoved at every weight tested.

The contrast is cleanest when task health is held constant rather than leakage, since MMD only reaches comparable leakage by destroying the classifier:

Gradient reversal reduces leakage $0.970 \rightarrow 0.763$ **at no cost in accuracy** ($0.845 \rightarrow 0.845$) and collapses all five detectors to chance. MMD reduces leakage $0.967 \rightarrow 0.900$ at a similarly negligible cost ($0.779 \rightarrow 0.730$) and collapses nothing (Mahalanobis $0.813 \rightarrow 0.716$). To go further, MMD must destroy the task.

The calibration signatures differ in the same direction. As MMD degrades, its out-of-domain ECE *falls* ($0.344 \rightarrow 0.142$): the model becomes less confident as it becomes less accurate, which is the appropriate response. Under gradient reversal out-of-domain ECE triples while accuracy falls. The two objectives move leakage the same way and fail in opposite manners — one degrades honestly, the other is confidently wrong.

We do not claim MMD would be safe at the invariance level gradient reversal reaches; it never got there. What the data establish is that at the leakage levels MMD does reach, it shows none of the failure, while gradient reversal at the same levels has already collapsed.

Table 3: Single-encoder DANN on the same data and splits. ID accuracy is plain accuracy; leakage is balanced accuracy, floor 0.5. All five detectors collapse at the smallest non-zero weight, and four of five fall below chance by $\lambda_{\text{adv}} = 4$.

λ_{adv}	n	Leakage	Maha.	k -NN	MSP	cosine	Energy	ID acc	X-dom
0	5	0.970	0.700	0.944	0.868	0.884	0.911	0.845	0.390
0.25	3	0.763	0.525	0.528	0.515	0.509	0.507	0.845	0.333
0.5	3	0.795	0.510	0.517	0.501	0.493	0.507	0.844	0.328
1	3	0.795	0.548	0.521	0.462	0.482	0.448	0.840	0.334
2	5	0.807	0.509	0.516	0.479	0.468	0.470	0.839	0.325
4	3	0.798	0.502	0.486	0.471	0.454	0.449	0.835	0.320
8	3	0.787	0.520	0.532	0.519	0.490	0.500	0.832	0.321

4.7 It is not specific to the factorized architecture

All results above use a dual-encoder model. Table 3 repeats the sweep with plain single-encoder domain-adversarial training: one backbone, one linear classifier, a gradient reversal layer and a domain head, with no context branch and no orthogonality term.

The cliff reproduces. Every detector collapses at $\lambda_{\text{adv}} = 0.25$ — k -NN $0.944 \rightarrow 0.528$, energy $0.911 \rightarrow 0.507$, cosine $0.884 \rightarrow 0.509$, MSP $0.868 \rightarrow 0.515$ — and by $\lambda_{\text{adv}} = 4$ four of the five sit below chance, cosine most clearly at 0.454 ± 0.014 . In-distribution accuracy is flat throughout ($0.845 \rightarrow 0.832$) and cross-domain accuracy falls from 0.390 to a plateau near 0.32. The invisibility, the collapse and the inversion are all properties of domain-adversarial training rather than of the factorized architecture.

One difference is informative. The single-encoder model plateaus at leakage 0.76–0.81 and never reaches the 0.553 the dual-encoder attains; more adversarial pressure buys no further invariance. The dual-encoder has somewhere to put the discarded information — a domain probe on its context branch reads > 0.9999 — while the single-encoder must destroy it. The architecture that achieves deeper invariance also suffers the larger collapse (0.413 versus 0.509 at $\lambda_{\text{adv}} = 2$), which is the dose–response relationship seen from a second direction.

4.8 What the phenomenon is not

Table 4 lists six candidate accounts, each tested against a prediction fixed in advance, and each refuted, together with a pre-registered synthetic reproduction that failed. We report them prominently because they are the reason the remaining claims are credible, and because delimiting a phenomenon is useful to whoever eventually explains it. We do not have a mechanism.

4.9 A domain monitor is cheap, and semantic novelty is unaffected

A linear domain head on a frozen ImageNet ResNet-50 that has never seen this data reaches AUROC 0.998; on the trained baseline, 0.997; on the trained EfficientNet-B3 control, 0.991. Detecting the acquisition domain is easy from almost any representation, so retaining a monitor is not in itself difficult. What distinguishes the dual-encoder’s context branch is *concentration*: a single principal component carries 84.5% of its variance and already separates the domains (AUROC > 0.9999 , two discordant pairs in 11.6M), against 0.55–0.70 at $k = 1$ for every other representation tested.

The damage is specific to covariate shift. With the domain held constant and diagnostic classes withheld from training instead, both branches land between 0.6 and 0.8 and the ordering depends on which class is withheld (DF and VASC: lesion 0.648, context 0.641, baseline 0.687; SCC: lesion 0.806, context 0.607, baseline 0.684). No near/far structure organises the difference (Pearson -0.12 against a centroid-similarity index). We report this as a null result.

Table 4: Candidate explanations tested and refuted. Each was a prediction specified before its test.

Explanation	Test	Result
Memorisation of the adversarial images	Hold PAD out of the adversary at patient level	<code>pad_heldout</code> 0.427 vs <code>pad_adv</code> 0.407; unchanged
Class composition	Restrict ID to the six shared classes; reweight ID to the OOD class mix	0.414 vs 0.409; reweighting <i>strengthens</i> the inversion to 0.240
Domain specificity	Evaluate on Fitzpatrick17k, never in $\mathcal{L}_{\text{adv}}$	0.399; a non-adversarial baseline reaches 0.994 on the same images
Dimensional collapse	Participation ratio and eigenspectrum across λ_{adv}	4.879 $\rightarrow$ 4.500 while leakage 0.914 $\rightarrow$ 0.553; decoupled
Latent compression	Latent variance and distance to centroid across λ_{adv}	OOD is already $4.5\times$ tighter at $\lambda_{\text{adv}} = 0$; the adversary <i>decompresses</i> it
Tail asymmetry	Recompute AUROC after truncating the pooled upper tail	0.412–0.424 versus 0.427; medians invert
Synthetic reproduction	Pre-registered toy model with heavy-tailed ID structure, grid fixed before fitting	No inversion ($\lambda_{\text{adv}}=0$: 0.693; $\lambda_{\text{adv}}=2$: 0.786)

5 Discussion

5.1 The deployment pathway

The results compose into a concrete failure sequence. A model is trained with invariance to acquisition domain, as recommended, and deployed at a new site. On images from that site the model increasingly predicts the benign majority class, with actinic keratosis recall falling from 0.244 to 0.016 (49 of 202 images correctly recalled, then 3). It does so with higher confidence than on its own test data. The OOD monitor that should flag those images for human review scores them as more routine than familiar ones. And the developer’s evaluation shows in-distribution accuracy holding, in-distribution calibration unchanged, and the leakage metric improved exactly as intended.

5.2 The danger zone is where the method succeeds

Risk scales with *achieved* invariance, not with the nominal weight or with the presence of an adversarial term. In a separate setting we ran the same objective with an adversary that failed to train; leakage stayed at 0.94 across four orders of magnitude of λ_{adv} , and the safety failure did not appear either. A weak adversary produces neither debiasing nor harm. This matters practically: many deployed configurations may be too weak to induce invariance, and are correspondingly harmless.

5.3 Tolerance without generalization

Full-spectrum OOD detection [6] prescribes that covariate-shifted in-distribution inputs be accepted rather than flagged. Our models achieve that prescription and are unsafe because of it. By $\lambda_{\text{adv}} = 2$ they accept out-of-domain images more readily than their own test data, while classifying them at 0.249 balanced accuracy against a floor of 0.167 and reporting higher confidence than in distribution.

Acceptance and competence are separate properties, and current protocols evaluate them separately or not at all. We suggest that any claim of tolerance to covariate shift be accompanied

by cross-domain accuracy on the same inputs, since tolerance of inputs a model cannot classify is a failure mode rather than a success.

5.4 Two routes to invariance are not interchangeable

Wald et al. [9] showed that enforcing multi-domain calibration produces invariance. We find the converse fails: enforcing invariance adversarially destroys out-of-domain calibration, driving ECE from 0.247 to 0.746. The property that one route uses as its foundation is the property the other route removes.

This suggests a concrete design preference. Where invariance is the goal, calibration-based objectives may be preferable to adversarial ones, because they arrive at invariance through a constraint that keeps the out-of-domain uncertainty signal intact rather than destroying it. We have not tested this directly and offer it as a hypothesis the present results motivate.

5.5 Recommendation

Report OOD detection and out-of-domain calibration **as a function of achieved invariance**, not at a single operating point; report cross-domain accuracy alongside any claim of tolerance to covariate shift; and do not use a separate dataset as the OOD set without saying that this measures covariate shift rather than semantic novelty. Practitioners tuning λ_{adv} already train at multiple values, so this is a change in what is reported rather than additional work. It converts an invisible failure into a visible number.

5.6 An open mechanism

Six accounts were tested and refuted and a pre-registered synthetic reproduction failed. We do not know why the inversion occurs. We note only that it is a bulk location shift, survives shrinkage estimation, is not mediated by linear domain-decodability, and appears on domains the adversary never saw. We report the eliminations so that the search space is smaller for whoever resolves it.

6 Limitations

Absolute cross-domain performance is poor for every model here. Plain accuracy on `pad_heldout` is 0.245 for the factorized model and 0.391 for the baseline, against a 0.385 majority rate. No model in this study is deployable. Our claim concerns the *effect of λ_{adv}* within a controlled intervention, not the deployability of any particular model, and the deployment pathway in the discussion should be read as a mechanism of harm rather than as a description of a fielded system.

One domain pair, one modality contrast. ISIC is dermoscopy; PAD-UFES and Fitzpatrick17k are clinical photography. The construct we call domain is confounded with imaging modality, and Fitzpatrick17k does not resolve this because it is also clinical photography. Generality beyond dermatology is untested.

Architecture scope. All results are obtained with a dual-encoder factorization model. The λ_{adv} sweep is a controlled intervention *within* that architecture; whether single-encoder domain-adversarial training behaves identically is not established here. An attempt to replicate on Camelyon17 was **inconclusive**: the adversary head never trained (cross-entropy $\ln 3$ from the first epoch at every λ_{adv} , while a post-hoc probe read 0.94), despite a gradient graph verified correct (head cosine +1.000, encoder −1.000). **We report no Camelyon17 result and make no claim that Camelyon17 resists invariance.** We record the diagnostic — an adversary failing to train under a verified-correct reversal — for future work.

Detector non-independence. The five scoring rules are functions of the same 16-dimensional latent. The backbone analysis provides the independent replication.

Calibration arithmetic. Out-of-domain ECE is close to mean confidence minus accuracy in this setting; the confidence, accuracy and ECE results are one phenomenon viewed three ways.

Seeds. Five seeds at the endpoints, three at interior points. Effect sizes are large relative to seed variance, but formal testing across seven points at $n = 3$ is not claimed.

Splits. ISIC splits are image-level, not patient-level; some patients may contribute images to both partitions. Because in-distribution accuracy is compared *across* λ_{adv} *under an identical split*, any inflation applies equally to every row and does not threaten the reported pattern, though absolute values may be optimistic. PAD splits are patient-level.

Third-domain coverage. 3,887 of 16,577 Fitzpatrick17k images were obtained; the set is web-scraped under mixed copyright, used for evaluation only, and not redistributed.

7 Conclusion

Domain-adversarial invariance training, applied successfully, leaves a skin lesion classifier confidently wrong on images from clinics it has never seen and removes its ability to flag them. In-distribution accuracy, in-distribution calibration and the domain leakage probe all continue to look correct while this happens. The model achieves the tolerance of covariate shift that current OOD taxonomies prescribe, without the generalization that would make that tolerance safe. We could not explain the inversion after refuting six candidate accounts, but explaining it is not required to act on it: reporting OOD detection, out-of-domain calibration and cross-domain accuracy as functions of achieved invariance costs almost nothing and makes the failure visible.

Data and code availability

[PENDING: repository link]. ISIC 2019 CC BY-NC 4.0; PAD-UFES-20 CC BY 4.0; Fitzpatrick17k annotations CC BY-NC-SA 3.0 with images used for evaluation only and not redistributed.

Declaration of competing interest

The authors declare no competing interests.

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